

A Web-Based Service Management System with Data Analytics using Machine Learning Algorithms for Shoelotskey Villamor-Pasay

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March 26, 2026

Chapter 1 : Introduction

Research Gap	Research Problem	Research Questions
Past studies show that web analytics, digital records, machine learning, and software testing standards are very useful. However, three main gaps still exist. First, there is a context gap because most research focuses on big companies instead of small local shops like Shoelotskey. Second, there is an integration gap because past studies look at these tools separately instead of combining them into one single system. Third, there is an evaluation gap because standard testing methods are rarely used for systems made for small businesses. Because of this, no past study has combined all these features into one web-based system made specifically for a Philippine shoe repair shop. This directly shows why developing the Shoelotskey system is needed.	The researchers identify that the Shoelotskey operates using traditional manual logbooks to track customer orders and shoe conditions. This manual approach is prone to human error, data loss, lack of backup, and slow retrieval of information. Furthermore, the business lacks a proper method for estimating service completion dates and often relies on mental guesswork, which can lead to missed deadlines and customer frustration. Finally, the absence of a centralized data system prevents the owner from having a clear overall view of the business, making it difficult to identify which services are most profitable.	RQ1: How can a centralized order entry system be designed and developed to improve data organization, accuracy, and retrieval efficiency compared to manual logbook recording? RQ2: How can a machine learning model be implemented to predict service completion times using historical turnaround data? RQ3: How can business analytics features be integrated to generate meaningful insights on sales trends, service demand, and profitability? RQ4: To what extent does the developed system meet the ISO/IEC 25010 software quality standards in terms of functionality, data accuracy, prediction reliability, and user satisfaction among Shoelotskey staff and management? RQ5: What is the level of user acceptance of the Shoelotskey Service Management System among staff and management using the Technology Acceptance Model (TAM)?

Chapter 1 : Project Context

- **Existing System**
 - Manual and semi-digital service recording
 - Paper-based job order tracking
 - No centralized monitoring or predictive system
- **Gaps and Risks**
 - Delayed updates and non-centralized storage
 - Inaccurate reporting and limited visibility
 - Lack of data-driven predictive and analytical capabilities
 - Risk of inefficiency as demand increases
- **Need for Innovation and Standards Basis**
 - Centralized web-based service management system
 - Integration of data analytics and machine learning
 - Automated monitoring and real-time reporting
 - ISO/IEC 25010 Software Quality Model
 - Data-driven decision-making principles
 - Digital transformation best practices

Chapter 1 : Purpose and Description

Project Purpose

- To design, develop, and evaluate a web-based service management system enhanced with data analytics and machine learning that improves operational efficiency, predictive accuracy, and standards-based quality compliance for Shoelotskey Villamor–Pasay.

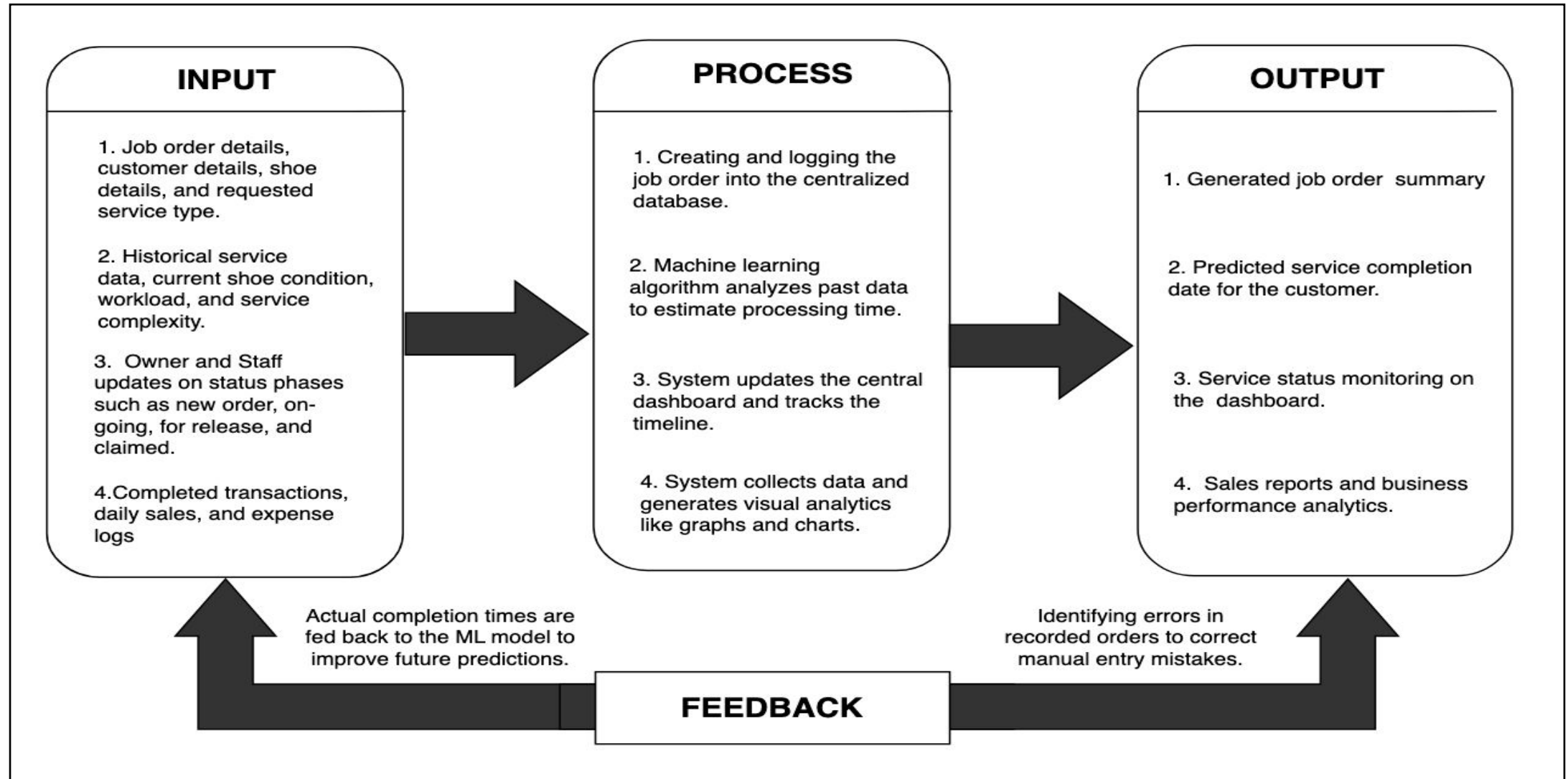
System Components

- Centralized order entry and service tracking
- Machine learning-based completion time prediction
- Business analytics dashboard with real-time reports
- Role-based access for owner and staff

Expected Benefits

- Improved accuracy and faster data retrieval
- Reliable service time estimation
- Better monitoring and decision-making
- Increased operational efficiency and scalability

Chapter 1 : Contextual Diagram



Chapter 1 : Objectives

General Objective

The researchers aims to design, develop, and evaluate a Web-Based Service Management System with Data Analytics using Machine Learning Algorithms for Shoelotskey Villamor–Pasay in order to automate service operations, improve data accuracy, enhance prediction of service completion dates, and support data-driven decision-making.

Research Questions	Research Objectives
RQ1	Design and develop a centralized order entry system that improves data organization, accuracy, and retrieval efficiency compared with manual logbook recording.
RQ2	Implement a machine learning model capable of predicting service completion times using historical turnaround data.
RQ3	Integrate business analytics features that generate meaningful insights on sales trends, service demand, and profitability.
RQ4	Evaluate the developed system using the ISO/IEC 25010 software quality standards regarding functionality, data accuracy, prediction reliability, and user satisfaction among Shoelotskey staff and management.
RQ5	Determine the level of user acceptance of the Shoelotskey Service Management System among staff and management using the Technology Acceptance Model (TAM).

Chapter 1: Scope and Limitations

Scope

- Implementation of a centralized web-based platform using a client–server architecture
- Focuses on security, ML driven qualitative analysis, usability, and ISO-aligned quality compliance
- Integration of descriptive data analytics for monitoring sales trends, service frequency, and workload distribution
- Application of a machine learning algorithm to estimate service completion time based on historical data
- System testing and evaluation based on recognized software quality attributes such as functionality, usability, reliability, performance efficiency, and security
- Deployment within the operational environment of Shoelotskey only

Limitations

- System is customized only for Shoelotskey and not directly applicable to other businesses without modification
- No mobile application version
- No customer-side access portal
- No online payment integration
- Machine learning prediction limited to available historical data
- Evaluation limited to selected internal users (owner and staff only)

Chapter 2: Review of Related Literature

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Chapter 3: Technical Background

- **Research & Development Methodology**
 - Adopted **Research and Development (R&D) approach**
 - Focused on designing, developing, and evaluating a functional system
 - Combined system creation with actual validation
- **Methodology Overview**
 - Adopted **Waterfall Software Development Life Cycle**
 - Sequential and structured development approach
 - Suitable for clearly defined institutional requirements
 - Ensures proper documentation

Waterfall Phases

1. Requirements Analysis
2. System Design
3. Implementation (Coding)
4. Testing
5. Deployment
6. Maintenance

Chapter 3: Technical Background

- **Analysis of Existing System & Gap**

- **Current System**

- Manual and paper-based recording of job orders
- Service tracking done through verbal updates or informal monitoring
- Sales computation performed manually
- No centralized database for customer and service records
- No real-time monitoring of service status and workload
- Decision-making based on experience rather than data

- **Identified Gaps**

- Absence of a centralized web-based service management platform
- Lack of real-time analytics for monitoring sales and service performance
- No predictive mechanism for estimating service completion time
- High risk of data redundancy, loss, and human error
- Limited operational visibility for owner and staff
- No structured digital workflow for tracking job orders and released items

Chapter 3: Technical Background

- **Proposed System Architecture**
- **Architecture Type**
 - Web-based Client–Server Model
 - Three-Layer Architecture:
 - Presentation Layer (User Interface)
 - Application Layer (Business Logic + Machine Learning Module)
 - Data Layer (Centralized Database)
- **Core Modules**
 - User & Access Management
 - Service Management
 - Job Order Management
 - Sales & Payment Recording
 - Service Operations Monitoring
 - Reports & Dashboard Analytics
 - Machine Learning Prediction

Chapter 3: Technical Background

- **Feasibility & Cost–Benefit Summary**
- **Feasibility Analysis**
 - **✓ Technical**
 - Uses open-source web and database technologies.
 - Runs on existing computers with internet access.
 - Machine learning limited to analytics-based predictions.
 - No specialized hardware required.
 - **✓ Operational**
 - Aligns with Shoelotskey’s daily workflow.
 - Replaces manual logs with centralized system.
 - Improves monitoring and reduces errors.
 - **✓ Economic**
 - Low development and hosting cost.
 - Reduces paper and admin expenses.
 - Long-term savings exceed initial cost.

Chapter 3: Technical Background

- **Feasibility & Cost–Benefit Summary**
- **Feasibility Analysis**
 - **✓ Schedule**
 - Achievable within 6-10 months.
 - Follows structured phases
 - Scope is manageable for capstone level.
 - **Cost-Benefits (3 years)**
 - Total Cost:
 - Total Benefit:
 - Net Benefit:
 - ROI:

Chapter 3: Technical Background

- **Security, Data Privacy & Technology Stack**
- **Security Measures**
 - Role-based authentication
 - HTTPS / SSL encrypted data transmission
 - Audit logging & activity monitoring
 - Compliance with RA 10173 (Data Privacy Act)
 - Restricted internal access
- **Technology Stack**
 - Web-based programming framework
 - Relational Database (RDBMS)
 - Python for prediction model (scikit-learn)
 - Web Server/Hosting

Chapter 3: Technical Background

- Procedural Flow

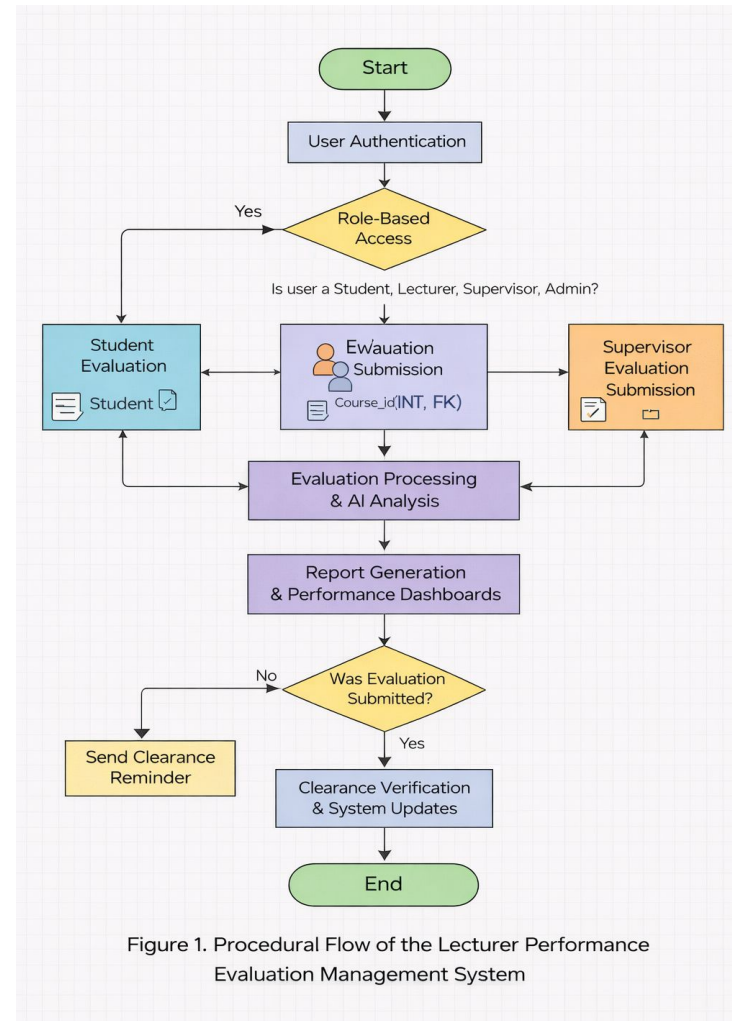
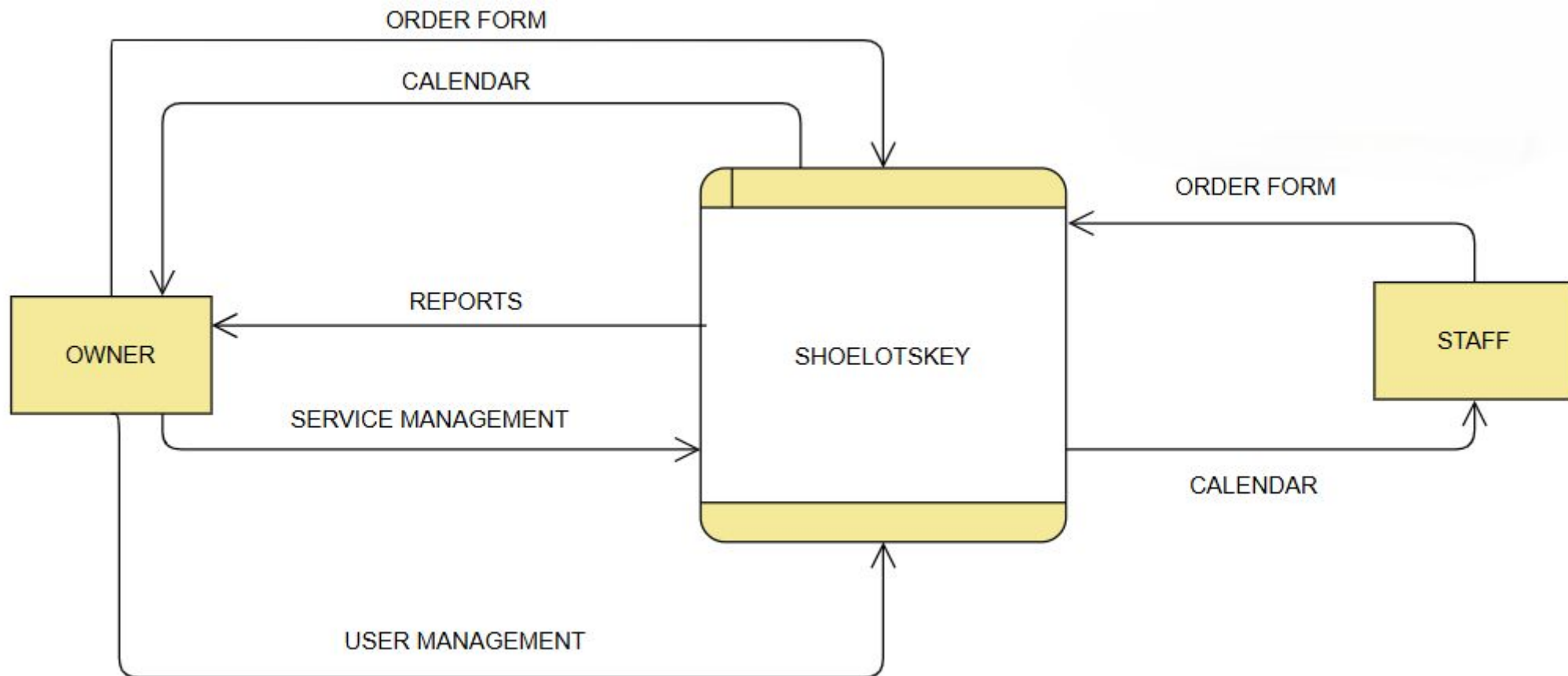


Figure 1. Procedural Flow of the Lecturer Performance Evaluation Management System

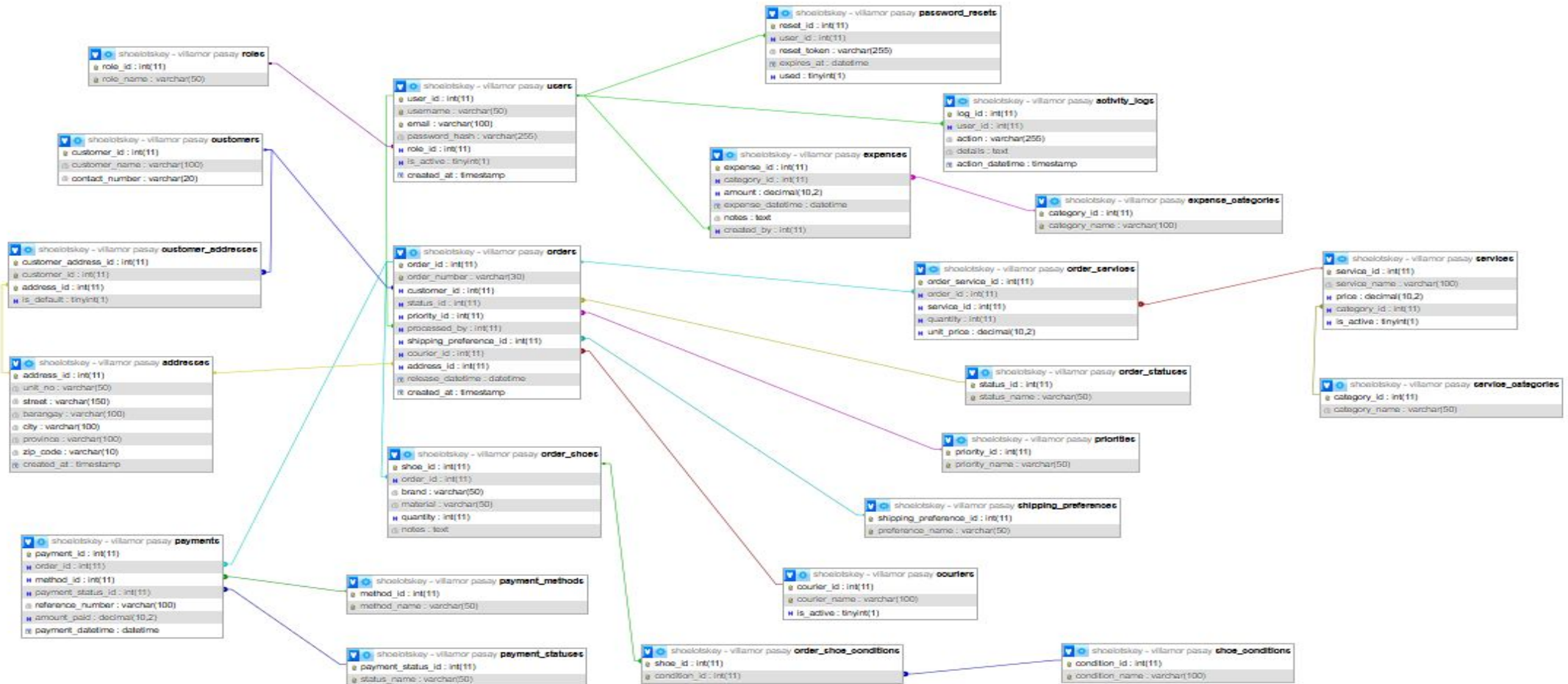
Chapter 3: Technical Background

- Data Flow Diagram Level 0



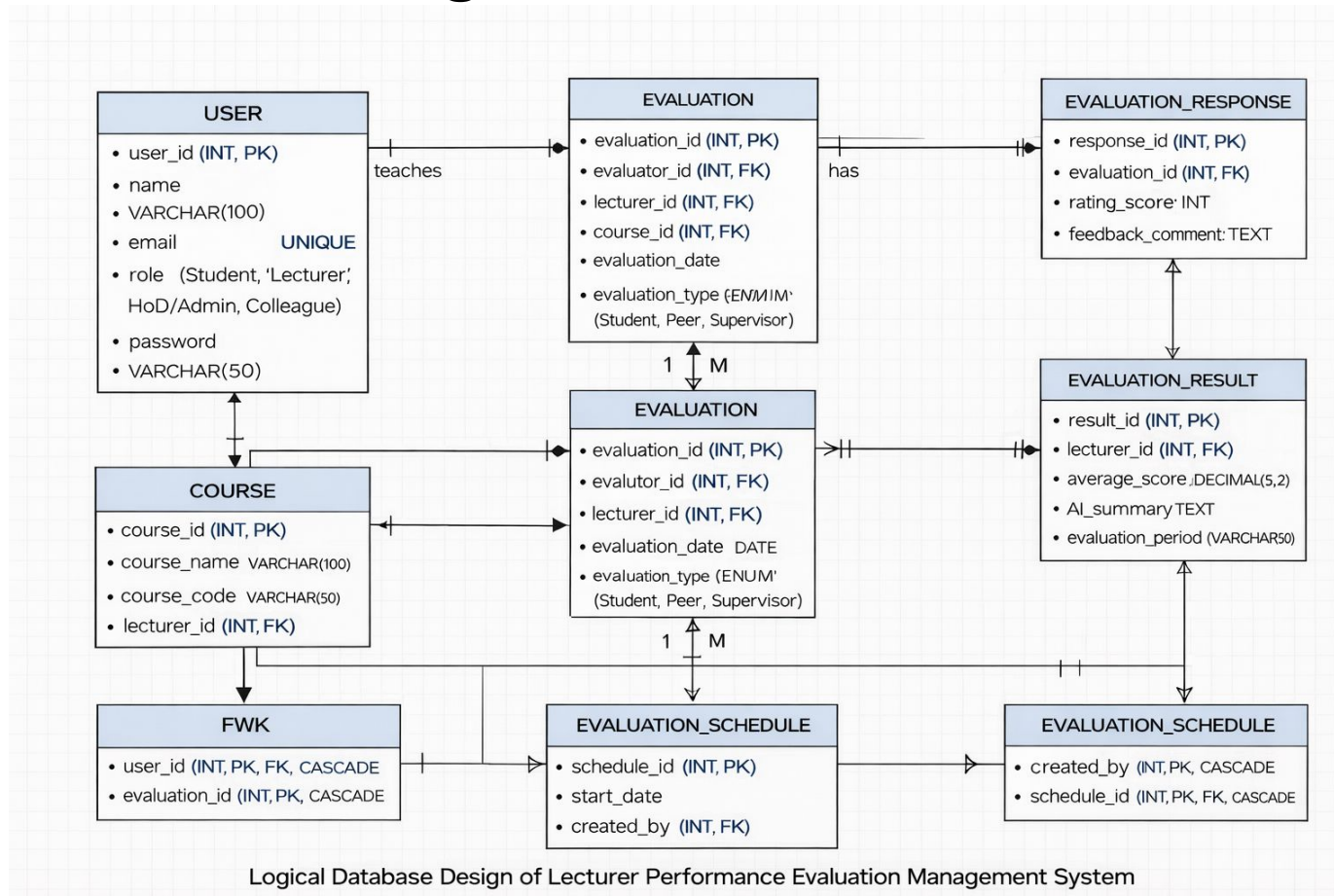
Chapter 3: Technical Background

- Entity Relationship Diagram



Chapter 3: Technical Background

• Logical Database Design



Chapter 4: Research Methodology

- **Research Design**

- This study utilized a **Developmental–Descriptive Research Design** to design, develop, and evaluate a Web-Based Service Management System with Data Analytics using Machine Learning Algorithms for Shoelotskey in Barangay 183 Villamor–Pasay. The developmental aspect focused on the systematic creation of the system as a functional technological solution to address the identified operational gaps, while the descriptive component was used to assess and describe the system's performance, usability, and effectiveness within the actual work environment.
- The study followed a structured **Software Development Life Cycle (SDLC)** using the **Waterfall Model**, ensuring a linear and organized progression through the phases of requirements gathering, system analysis, system design, development, testing, and deployment. This approach provided clear documentation, defined deliverables, and controlled implementation at each stage of the project. Overall, the research design ensured that the developed system was not only technically sound but also aligned with the operational needs of Shoelotskey and suitable for capstone-level evaluation and validation.

Chapter 4: Research Methodology

- Methodology Mapping

Research Objective	Data Source	Data Gathering Method	Statistical / Analytical Technique	Expected Output
1. Analyze the existing service workflow and identify operational gaps	Interview, Observation, Document Review	Requirements Analysis	Thematic Analysis	Documented System Requirements & Gap Analysis
2. Design the system architecture and database structure	Diagrams (DFD Level 0 and 1, Contextual, ERD, Procedural), Wireframe/UI	System Design	Logical System Modeling	System Blueprint & Database Schema
3. Develop the web-based service management modules	MySQL, Python, Figma, Github, Gemini Antigravity	System Development	Functional Testing	Working System Modules
4. Implement data analytics dashboard for monitoring	SQL Aggregation	Data Processing	Descriptive Statistics (Frequency, Trends)	Real-Time Operational Dashboard
5. Develop machine learning model for service completion prediction	Python (scikit-learn), Historical Service Data	Machine Learning Modeling	Regression Analysis, MAE (Mean Absolute Error)	Service Completion Time Prediction Model
6. Evaluate system quality and usability	System Evaluation	ISO/IEC 25010-based Survey, User Acceptance Testing	Mean, Percentage, Standard Deviation	System Evaluation & Validation Report

References

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References

Q & A

Thank You!